

Soldering the u-blox MAX-M10S GPS to the STEVAL-MKBOXPRO

MovementLogger firmware — UART4 wiring on the JP2 programming connector

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What you are doing

The MovementLogger firmware reads GPS over **UART4** on MCU pins **PA0 (TX)** and **PA1 (RX)**, at **38400 baud, 8 data bits, no parity, 1 stop bit (8N1)**, 3.3 V logic. The receiver is a separate **u-blox MAX-M10S** module — it is *not* on the STEVAL-MKBOXPRO. This sheet covers wiring that module to the box by hand.

The STEVAL-MKBOXPRO has no STMod+ connector. On the SensorTile.box PRO core board, UART4 (PA0/PA1) is brought out only on **JP2, the programming connector** — the same 14-pin header the STLINK-V3 plugs into. That is where you solder.

Sources: ST *STEVAL-MKBOXPRO schematic* Rev 3 (Figure 3, *Sensortilebox_pro_MCU*), ST *UM3133* Rev 7 §1.8.6 / §2.3, and the firmware (`Src/gps.c`, `Inc/config.h`).

Connector

Item	Value
Reference	JP2 (“Prog Connector” / programming connector)
Part	Samtec FTSH-107-01-L-D
Layout	2 × 7 = 14 pins, 1.27 mm (0.05”) pitch , shrouded
Also exposes	SWD/JTAG + NRST (leave those alone)

This is fine-pitch surface work. You need: a fine conical tip, flux, magnification, and thin wire — **30 AWG** enamelled / wire-wrap (≤ 0.5 mm). Tweezers and a steady bench.

JP2 pinout (from schematic Figure 3)

JP2 (pin 1 = square pad / silk triangle)	
odd column	even column
+-----+	
1 -> 1V8 rail (ref)	2 -> (no connect)
3 -> (do not use)	4 -> SWDIO / JTMS
5 -> (do not use)	6 -> SWCLK / JTCK
7 -> GND	8 -> JTD0 / SWO
9 -> (do not use)	10 -> JTDI
11 -> GND	12 -> NRST
13 -> UART4_RX (PA1)	14 -> UART4_TX (PA0)
+-----+	

- **Pin 13 = UART4_RX = MCU PA1** — the MCU’s *input*.

- **Pin 14 = UART4_TX = MCU PA0** — the MCU’s *output*.
- **Pin 7 and pin 11 = GND** (the only two GND pins confirmed by the schematic’s junction dots — use one of these for the GPS ground).
- **Pin 1 ties to the on-board 1.8 V rail** (the JTAG pull-up reference; R52/R54/R55 are 10 k to 1V8). It is **1.8 V, not 3.3 V**, and it is **not** a power source — never connect GPS VCC here. Take GPS 3.3 V from JP4 (see next section).
- Pins **4, 6, 8, 10, 12** are the debug bus. A solder bridge from pin 14 to pin 12, or onto SWDIO/SWCLK, can brick debug access — keep these clean.
- Pins **3, 5, 9** carry no signal you need — leave them unconnected.

Verify pin 1’s position against the square pad / silkscreen triangle on *your* board before counting, and **meter-confirm** every pin (continuity to MCU PA0/PA1 and to battery –) before soldering. Treat this table as a guide, not gospel — connector orientation can differ between board builds.

Wiring (note the cross-over)

UART is always TX-to-RX. The GPS and the MCU each have their own TX/RX, so the data lines cross:

GPS module pin	→	JP2 pin	Net (MCU)
TX (GPS out)	→	pin 13	UART4_RX (PA1)
RX (GPS in)	→	pin 14	UART4_TX (PA0)
GND	→	JP2 pin 7 or pin 11 (meter-verified)	GND
VCC 3.3 V	→	board 3 V rail — see below	—

GPS power: 3.3 V from the board rail

Per UM3133 §1.8.6 the **JP4 extension-board connector** (Multicomp MC-SVT1-S07-G, 7-pin) carries the board supply and **GND**, with a **domain switch selectable 1.8 V / 3 V**. To power the GPS from the board rail:

1. Set the extension-board supply domain to **3 V** (not 1.8 V).
2. With the board powered, **measure with a multimeter** that the pin reads **3.25–3.40 V** before you connect the GPS VCC to it.
3. Take **GND** from the same JP4 connector (or a JP2 GND pin) — the GPS ground and the box ground must be common.

The GPS will be powered and unpowered together with the box. There is **no graceful shutdown** — the Hall-sensor magnet cuts the whole supply rail at once (firmware design F-PWR-5). This is expected; the firmware and on-SD format tolerate a sudden cut.

Safety — read before the iron is hot

- **3.3 V logic only.** PA0/PA1 are **not 5 V tolerant**. A 5 V GPS breakout’s TX into PA1 will destroy the MCU. If your module is a 5 V board, level-shift or use a 3.3 V-native MAX-M10S carrier.
- **Never bridge to the SWD pins** (4/6/8/10/12).
- Keep each pad heated < 2 s; FTSH-107 pads lift if cooked.
- **Strain-relieve** the wires (Kapton tape + a dab of hot glue) close to JP2. A tugged wire rips the pad off the PCB.

Procedure

1. **Power down completely:** power switch S1 off, USB unplugged, LiPo disconnected.
2. **Locate JP2** — the 14-pin fine-pitch header by the MCU (the STLINK programming connector). Identify pin 1.
3. **Continuity-check before soldering** (meter, board off):
 - JP2 pin 13 ↔ MCU **PA1**
 - JP2 pin 14 ↔ MCU **PA0**
 - candidate GND pin ↔ battery (–) Confirm, don't assume — connector orientation differs between board builds.
4. **Pre-tin** the three JP2 pads and the wire ends with flux.
5. **Solder the data wires:** pin 13 ← GPS TX, pin 14 ← GPS RX. Then **GND**.
6. **GPS VCC last:** with the board powered, confirm **3.3 V** at the JP4 3 V pin, power down again, solder the VCC wire.
7. **Inspect under magnification** for bridges — especially pin 14 to pin 12 and onto the SWD pins.
8. **Strain-relieve** all four wires.

Bring-up test

1. Insert the SD card, take the box **outdoors with clear sky**, power on.
2. The firmware sends the u-blox UBX config on boot, then expects 10 Hz NMEA. First fix is typically reached within ~**30–60 s** cold.
3. Confirm a fix by either:
 - GpsNNN.csv on the SD card gaining rows (one per fix), or
 - the BLE SensorStream gps_valid flag going set, or
 - (antenna-test build, make GPS_ANTENNA_BEEP=1) the buzzer beeping N times every 10 s with N = satellites.

No fix after several minutes outdoors?

In order of likelihood:

1. **Swap the two data wires** (pin 13 ↔ pin 14). TX/RX reversed is the single most common wiring mistake and swapping is harmless to try.
2. Re-check **GND** continuity (cold/again) and that GPS VCC really is 3.3 V *at the module* under load.
3. Confirm the module is a 3.3 V part and actually powered (most MAX-M10S carriers have a power LED).
4. Give it a longer cold-start with an unobstructed sky view; indoors it will never fix.